

Digital Design and Computer Organisation Laboratory

UE23CS251A

3rd Semester, Academic Year 2024-25

Date:24/10/2024

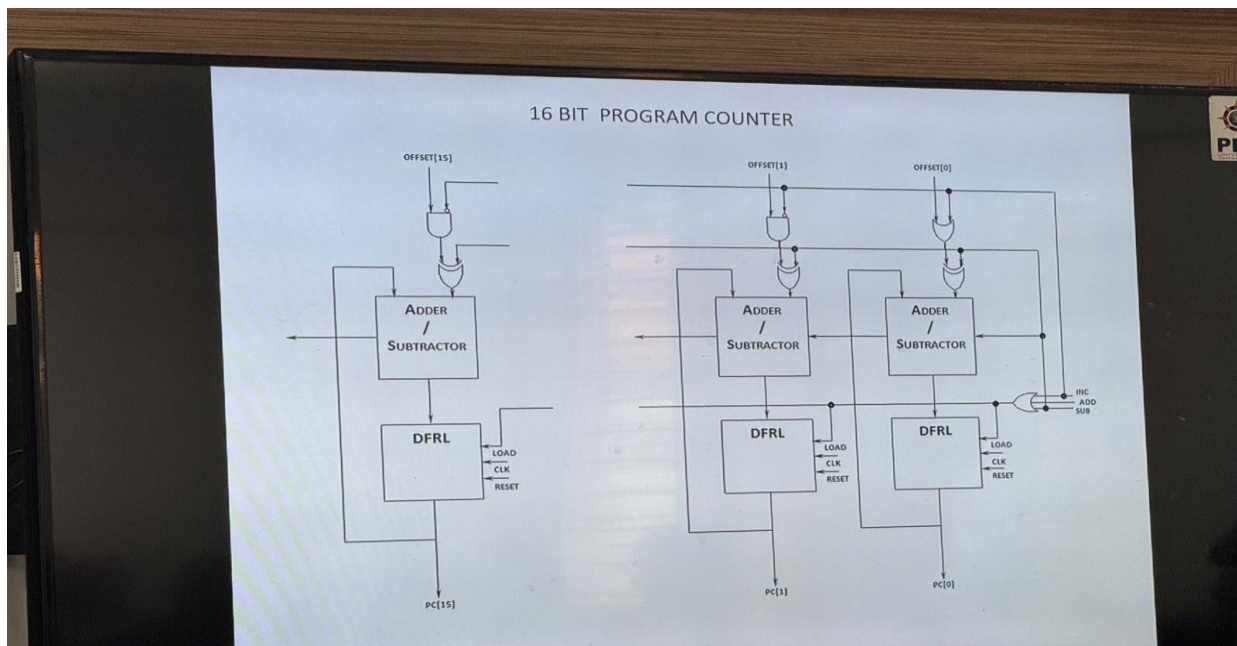
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Week# 7

Program Number: 1

TITLE: 16 Bit Program Counter

WRITE A VERILOG PROGRAM TO MODEL A 16 Bit PC Program Counter. GENERATE THE VVP OUTPUT AND SIMULATION WAVEFORM USING GTKWAVE. VERIFY THE OUTPUT AND WAVEFORM.



16 Bit PC Program Counter

lib.v file

```
week7 > ≡ lib.v U X
1 module invert (input wire i, output wire o);
2     assign o = !i;
3 endmodule
4
5 module and2 (input wire i0, i1, output wire o);
6     assign o = i0 & i1;
7 endmodule
8
9 module or2 (input wire i0, i1, output wire o);
10    assign o = i0 | i1;
11 endmodule
12
13 module xor2 (input wire i0, i1, output wire o);
14    assign o = i0 ^ i1;
15 endmodule
16
17 module nand2 (input wire i0, i1, output wire o);
18    wire t;
19    and2 and2_0 (i0, i1, t);
20    invert invert_0 (t, o);
21 endmodule
22
23 module nor2 (input wire i0, i1, output wire o);
24    wire t;
25    or2 or2_0 (i0, i1, t);
26    invert invert_0 (t, o);
27 endmodule
28
29 module xnor2 (input wire i0, i1, output wire o);
30    wire t;
31    xor2 xor2_0 (i0, i1, t);
32    invert invert_0 (t, o);
33 endmodule
34
35 module and3 (input wire i0, i1, i2, output wire o);
36    wire t;
37    and2 and2_0 (i0, i1, t);
38    and2 and2_1 (t, i2, o);
39 endmodule
40
41 module or3 (input wire i0, i1, i2, output wire o);
42    wire t;
43    or2 or2_0 (i0, i1, t);
44    or2 or2_1 (t, i2, o);
45 endmodule
46
47 module nor3 (input wire i0, i1, i2, output wire o);
48    wire t;
49    or2 or2_0 (i0, i1, t);
50    nor2 nor2_0 (t, i2, o);
51 endmodule
52
53 module nand3 (input wire i0, i1, i2, output wire o);
54    wire t;
55    and2 and2_0 (i0, i1, t);
56    nand2 nand2_1 (t, i2, o);
57 endmodule
58
59 module xor3 (input wire i0, i1, i2, output wire o);
60    wire t;
61    xor2 xor2_0 (i0, i1, t);
62    xor2 xor2_1 (t, i2, o);
63 endmodule

week7 > ≡ lib.v U X
64 endmodule
65
66 module xnor3 (input wire i0, i1, i2, output wire o);
67    wire t;
68    xor2 xor2_0 (i0, i1, t);
69    xnor2 xnor2_0 (t, i2, o);
70 endmodule
71
72 module mux2 (input wire i0, i1, j, output wire o);
73    assign o = (j==0)?i0:i1;
74 endmodule
75
76 module mux4 (input wire [0:3] i, input wire j1, j0, output wire o);
77    wire t0, t1;
78    mux2 mux2_0 (i[0], i[1], j1, t0);
79    mux2 mux2_1 (i[2], i[3], j1, t1);
80    mux2 mux2_2 (t0, t1, j0, o);
81 endmodule
82
83 module mux8 (input wire [0:7] i, input wire j2, j1, j0, output wire o);
84    wire t0, t1;
85    mux4 mux4_0 (i[0:3], j2, j1, t0);
86    mux4 mux4_1 (i[4:7], j2, j1, t1);
87    mux2 mux2_0 (t0, t1, j0, o);
88 endmodule
89
90 module demux2 (input wire i, j, output wire o0, o1);
91    assign o0 = (j==0)?i:1'b0;
92    assign o1 = (j==1)?i:1'b0;
93 endmodule
94
95 module demux4 (input wire i, j1, j0, output wire [0:3] o);
96    wire t0, t1;
97    demux2 demux2_0 (i, j1, t0, t1);
98    demux2 demux2_1 (t0, j0, o[0], o[1]);
99    demux2 demux2_2 (t1, j0, o[2], o[3]);
100 endmodule
101
102 module demux8 (input wire i, j2, j1, j0, output wire [0:7] o);
103    wire t0, t1;
104    demux2 demux2_0 (i, j2, t0, t1);
105    demux4 demux4_0 (t0, j1, j0, o[0:3]);
106    demux4 demux4_1 (t1, j1, j0, o[4:7]);
107 endmodule
108
109 module df (input wire clk, in, output wire out);
110    reg df_out;
111    always@(posedge clk) df_out <= in;
112    assign out = df_out;
113 endmodule
114
115 module dfr (input wire clk, reset, in, output wire out);
116    wire reset_, df_in;
117    invert invert_0 (reset, reset_);
118    and2 and2_0 (in, reset_, df_in);
119    df df_0 (clk, df_in, out);
120 endmodule
121
122 module dfr1 (input wire clk, reset, load, in, output wire out);
123    wire _in;
124    mux2 mux2_0(out, in, load, _in);
125    dfr dfr_1(clk, reset, _in, out);
126 endmodule
```

pc.v file

```
lib.v U pc.v U X tb_pc.v U
week7 > pc.v
1 module fa (input wire i0, i1, cin, output wire sum, cout);
2     wire t0, t1, t2;
3
4     xor3_i0 (i0, i1, cin, sum);
5
6     and2_i1 (i0, i1, t0);
7
8     and2_i2 (i1, cin, t1);
9
10    and2_i3 (cin, i0, t2);
11
12    or3_i4 (t0, t1, t2, cout);
13
14 endmodule
15
16 module addsub (input wire addsub, i0, i1, cin, output wire sumdiff, cout);
17
18     wire t;
19
20     fa_i0 (i0, t, cin, sumdiff, cout);
21
22     xor2_i1 (i1, addsub, t);
23
24 endmodule
25
26 module pc_slice (input wire clk, reset, cin, load, inc, sub, offset,
27     output wire cout, pc);
28
29     wire in, inc_;
30
31     invert invert_0 (inc, inc_);
32
33     and2_and2_0 (offset, inc_, t);
34
35     addsub addsub_0 (sub, pc, t, cin, in, cout);
36
37     dfr1 dfr1_0 (clk, reset, load, in, pc);
38
39 endmodule
lib.v U pc.v U tb_pc.v U
week7 > pc.v
28 endmodule
40
41 module pc_slice0 (input wire clk, reset, cin, load, inc, sub, offset, output wire cout, pc);
42
43     wire in;
44
45     or2_or2_0 (offset, inc, t);
46
47     addsub addsub_0 (sub, pc, t, cin, in, cout);
48
49     dfr1 dfr1_0 (clk, reset, load, in, pc);
50
51 endmodule
52
53 module pc (input wire clk, reset, inc, add, sub, input wire [15:0] offset, output wire [15:0] pc);
54
55     input wire load;
56     input wire [15:0] c;
57
58     or3_or3_0 (inc, add, sub, load);
59
60     pc_slice0 pc_slice_0 (clk, reset, sub, load, inc, sub, offset[0], c[0], pc[0]);
61     pc_slice pc_slice_1 (clk, reset, c[0], load, inc, sub, offset[1], c[1], pc[1]);
62     pc_slice pc_slice_2 (clk, reset, c[1], load, inc, sub, offset[2], c[2], pc[2]);
63     pc_slice pc_slice_3 (clk, reset, c[2], load, inc, sub, offset[3], c[3], pc[3]);
64     pc_slice pc_slice_4 (clk, reset, c[3], load, inc, sub, offset[4], c[4], pc[4]);
65
66     pc_slice pc_slice_5 (clk, reset, c[4], load, inc, sub, offset[5], c[5], pc[5]);
67
68     pc_slice pc_slice_6 (clk, reset, c[5], load, inc, sub, offset[6], c[6], pc[6]);
69
70     pc_slice pc_slice_7 (clk, reset, c[6], load, inc, sub, offset[7], c[7], pc[7]);
71
72     pc_slice pc_slice_8 (clk, reset, c[7], load, inc, sub, offset[8], c[8], pc[8]);
73
74     pc_slice pc_slice_9 (clk, reset, c[8], load, inc, sub, offset[9], c[9], pc[9]);
75
76     pc_slice pc_slice_10 (clk, reset, c[9], load, inc, sub, offset[10], c[10], pc[10]);
77
78     pc_slice pc_slice_11 (clk, reset, c[10], load, inc, sub, offset[11], c[11], pc[11]);
79
80     pc_slice pc_slice_12 (clk, reset, c[11], load, inc, sub, offset[12], c[12], pc[12]);
81     pc_slice pc_slice_13 (clk, reset, c[12], load, inc, sub, offset[13], c[13], pc[13]);
82     pc_slice pc_slice_14 (clk, reset, c[13], load, inc, sub, offset[14], c[14], pc[14]);
83     pc_slice pc_slice_15 (clk, reset, c[14], load, inc, sub, offset[15], c[15], pc[15]);
84
85 endmodule
```

tb_pc.v file

```
week7 > tb_pc.v
1  `timescale 1 ns / 100 ps
2  `define TESTVECS 5
3
4  module tb;
5      reg clk, reset, inc, add, sub;
6      reg [15:0] offset;
7      wire [15:0] pc;
8      reg [18:0] test_vecs [0:(`TESTVECS-1)];
9      integer i;
10     initial begin $dumpfile("tb_pc.vcd"); $dumpvars(0,tb); end
11     initial begin reset = 1'b1; #12.5 reset = 1'b0; end
12     initial clk = 1'b0; always #5 clk =~ clk;
13
14     initial begin
15         test_vecs[0][18] = 1'b1; test_vecs[0][17] = 1'b0; test_vecs[0][16] = 1'b0;
16         test_vecs[0][15:0] = 15'hxx;
17         test_vecs[1][18] = 1'b0; test_vecs[1][17] = 1'b1; test_vecs[1][16] = 1'b0;
18         test_vecs[1][15:0] = 15'ha5;
19         test_vecs[2][18] = 1'b0; test_vecs[2][17] = 1'b0; test_vecs[2][16] = 1'b0;
20         test_vecs[2][15:0] = 15'hxx;
21         test_vecs[3][18] = 1'b1; test_vecs[3][17] = 1'b0; test_vecs[3][16] = 1'b0;
22         test_vecs[3][15:0] = 15'hxx;
23         test_vecs[4][18] = 1'b0; test_vecs[4][17] = 1'b0; test_vecs[4][16] = 1'b1;
24         test_vecs[4][15:0] = 15'h14;
25     end
26
27     initial {inc, add, sub, offset} = 0;
28     pc <= pc_0 (clk, reset, inc, add, sub, offset, pc);
29     initial begin
30         #6 for(i=0;i<`TESTVECS;i=i+1)
31             begin #10 {inc, add, sub, offset}=test_vecs[i]; end
32         #100 $finish;
33     end
34
35     always@(reset or inc or add or sub )
36
37     $monitor("At time = %t, Reset= %b,inc=%b, add=%b,sub = %b,pc =%h ", $time,reset,inc,add,sub,pc);
38
39 endmodule
```

Verilog VVP Output Screen Shot:-

```
C:\Windows\System32\cmd.e  X  +  v
(c) Microsoft Corporation. All rights reserved.
C:\DDCO_IVERILOG\week7>iverilog -o test lib.v pc.v tb_pc.v
C:\DDCO_IVERILOG\week7>vvp test
VCD info: dumpfile tb_pc.vcd opened for output.
At time =          0, Reset= 1,inc=0, add=0,sub = 0,pc =xxxx
At time =         50, Reset= 1,inc=0, add=0,sub = 0,pc =0000
At time =        130, Reset= 0,inc=0, add=0,sub = 0,pc =0000
At time =        160, Reset= 0,inc=1, add=0,sub = 0,pc =0000
At time =        250, Reset= 0,inc=1, add=0,sub = 0,pc =0001
At time =        260, Reset= 0,inc=0, add=1,sub = 0,pc =0001
At time =        350, Reset= 0,inc=0, add=1,sub = 0,pc =00a6
At time =        360, Reset= 0,inc=0, add=0,sub = 0,pc =00a6
At time =        460, Reset= 0,inc=1, add=0,sub = 0,pc =00a6
At time =        550, Reset= 0,inc=1, add=0,sub = 0,pc =00a7
At time =        560, Reset= 0,inc=0, add=0,sub = 1,pc =00a7
At time =        650, Reset= 0,inc=0, add=0,sub = 1,pc =0093
At time =        750, Reset= 0,inc=0, add=0,sub = 1,pc =007f
At time =        850, Reset= 0,inc=0, add=0,sub = 1,pc =006b
At time =        950, Reset= 0,inc=0, add=0,sub = 1,pc =0057
At time =       1050, Reset= 0,inc=0, add=0,sub = 1,pc =0043
At time =       1150, Reset= 0,inc=0, add=0,sub = 1,pc =002f
At time =       1250, Reset= 0,inc=0, add=0,sub = 1,pc =001b
At time =       1350, Reset= 0,inc=0, add=0,sub = 1,pc =0007
At time =       1450, Reset= 0,inc=0, add=0,sub = 1,pc =ffff3
At time =       1550, Reset= 0,inc=0, add=0,sub = 1,pc =ffdf
tb_pc.v:32: $finish called at 1560 (100ps)
C:\DDCO_IVERILOG\week7>
```

GTKWAVE Screenshot:-

